



Programming 2 Project



Write Java Program to simulate online Store System. Any customer enter the store will assign a new shopping cart. The shopping cart is a collection of items that a customer uses to collect (items)things for purchase. A customer can add items to the cart, remove them, empty the cart, view the items in the cart, and end shopping and proceed to checkout.

The program must have a menu showing its services as shown below:

Store System Menu

1. Add a new Item to Store
2. Add a new Customer to Store
3. Add an item to Customer shopping cart
4. Remove an item from Customer shopping cart
5. View the items in Customer shopping cart
6. End shopping and go to checkout
7. Empty Customer shopping cart
8. Modify customer data
9. Exit the program.

The above services must do the following actions

1. Add a new Item to Store.

The system prompts to the user for adding a new item data. The item data contains the following items: -

- item number: holds an unique number for the item.
- - item name: holds the full name of the item.
- - item quantity: holds the quantity of the item.
- - item price: holds the price of the item

Add Item Operation :

Please enter the item number : 001

Please enter the item Name : Fifa 20 game

Please enter the quantity : 200

Please enter the price : 20.56

Enter the type of item (B)Book, (s)shoes, (G)Game ? : G

Item added Successfully

Do you want add another item (y/n)? y

Add Item Operation :

Please enter the item number : 002

Please enter the item Name : Calculus Book

Please enter the quantity : 100

Please enter the price : 95.2

Enter the type of item (B)Book, (s)shoes, (G)Game ? : B

Enter the Book title : introduction of Calculus Math

Enter the Book author name : Rami Attaallah

Item added Successfully

Do you want add another item (y/n)? n

2. Add a new Customer.

The system prompts to the system user for adding data for a new customer data.

The customer data contains the following items:

- customer No : holds an unique number of a customer.
- - customer name: holds a name of the course.
- - New Shopping cart will be create and added automatically for the new customer. The shopping cart will be empty.

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Add new Customer Operation :  
Please enter the customer number : 0011  
Please enter the customer Name : Rami Attaallah  
Customer added with empty shopping cart Successfully  
Do you want add another item (y/n)? n
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The System will repeat adding another customer , if the user answer yes for the question " Do you want add another customer (y/n)? ". if the user select no, the system will return to the main menu showing it's services.

3. Add an item to Customer shopping cart

The system asks the user about a wanted customer for adding an item to his shopping cart.

The system will allow the user to specify the customer by a customer number.

The system will search about the specified customer to show if he/she is exists or not exists.

If the customer is exist, The system will display a form displaying all items in the store, Then the system will ask the user to enter the items number and the required item quantity that the customer wants to add to his shopping cart.

For each input the system will check the required item quantity with the item quantity in the store to validate that the required quantity is available in the store for that item , if not that the system will alert the user and will not allow the user to add the item to shopping cart.

Add Item to Customer Shopping Cart Operation :

Please Enter customer number: 212

The customer is not exists, Do you want try again(y/n)? y

Please Enter customer number: 215

The customer no :215 ,The customer name : Rami Attaallah

Items in the store

>>> 1. 1401 Java Book. [3]

>>> 2. 1311 logo Game. [5]

>>> 3. 1444 Nike sport shoes [1]

>>> 0. Return to a main menu

Enter your choice item : 3

The item is [1444 Nike sport shoes [1]]

Enter the quantity you need : 2

Sorry the required quantity is not available, the available quantity is [1],try again!

Do you want add another item to shopping cart (y/n)? y

Enter your choice item : 3

The item is [1444 Nike sport shoes [1]]

Enter the quantity you need : 1

The item is [1444 Nike sport shoes [1]] adding to shopping cart is success

Do you want add another item to shopping cart (y/n)? y

Enter your item choice :1

The item is [1401 Java Book. [3]]

Enter the required quantity: 1

The item is [1401 Java Book. [1]] adding to shopping cart is success

Do you want add another item to shopping cart (y/n)? n

Sample input for this case:

4. Remove an item from Customer shopping cart

When the user select this option, the System will prompts the user to enter a customer number(id) and then the system will search about the customer.

If the customer is exists ,The system will show the customer information including items of his shopping cart and will prompt the user to enter the item choice to remove from the shopping cart.

Remove Item from Customer Shopping Cart Operation :

Enter customer number: **215**

The customer no :215 , The customer name : Rami attaallah.

The current items in the shopping cart.

>>> 1. [1444 Nike sport shoes [1]]

>>> 2. [1401 Java Book. [1]]

What you want to modify?

>>> R. Remove item form the shopping cart.

>>> M. Return to the main menu.

Enter your choice [R Remove, M main menu] : **R**

Enter Your item option number : **1**

The item [1444 Nike sport shoes [1]] is removed from shopping cart.

What you want to modify ?

>>> R. Remove item form the shopping cart.

>>> M. Return to the main menu.

Enter your choice [R Remove, M main menu] : **M**

5. View the items in Customer shopping cart

The System should display the items in shopping cart for a given customer.

View the items in Customer shopping cart Operation :

Enter customer number: 215

Item no	Item name	Quantity	Unit price	Total price
1401	Java Book	1	20	20
				Total price 20

Press (m/M) key to return to the main menu : m

6. End shopping and go to checkout

The System should display the items in shopping cart for a given customer and remove the customer from the store.

7. Empty Customer shopping cart

The System should remove all items in shopping cart for a given customer or assign new shopping cart for the customer.

8. Modify Customer Data

The system must modify the customer's data using the customer's number.

Modify Customer Data:

Enter customer number: 218

The customer no :218 „not found.

Do you want modify another item (y/n)? y

Enter customer number: 215

The customer no :215 , The customer name : Rami attaallah.

Please enter the customer number : 012

Please enter the customer Name : Ahmed Attaallah

Customer modified Successfully

Do you want modify another item (y/n)? n

back to main menu

Notes:

- This project is to be done individually (i.e. each student must submit his own project).
- Use all Object Oriented Concept Which You Learned in the Course.
- You must organize your code in classes and methods for example Customer class, Store class, ShoppingCart.
- No grades without discussions.
- ~~This Project Should be Submitted Before at Wednesday 29 / 4 / 2023 at 11:59 PM.~~
- Copying projects from friends or copying from the Internet or ChatGPT (**zero** mark for all copies).